

ROAD-SENSE — Bharat Road Health Network

Complete YEC 2026 Presentation & Preparation Handout



Core proposition: make road-condition evidence a by-product of journeys using ground sensing, NavIC/GNSS positioning, edge computing, multi-sensor fusion and road-health analytics.

1. Introduction

ROAD-SENSE is a proposed specialized road-health intelligence layer. It is not another turn-by-turn navigation system. The system collects road-level evidence during journeys and converts it into segment-level health, confidence, history and trend information.

2. WHO → WHAT → CONSTRAINT

WHO: commuters, fleets, school/institutional vehicles, logistics operators, citizens and road-maintenance teams.

WHAT: road defects are difficult to continuously observe, structure, verify and prioritize. CONSTRAINT: periodic/manual inspection, optional reporting and noisy individual observations.

3. System perspective

Six layers: Sense → Locate → Compute → Understand → Serve → Act. Sense uses camera/IMU/context sensors. Locate uses NavIC/GNSS. Compute uses ESP32 and an edge computer. Understand performs detection, sensor fusion and confidence estimation. Serve creates maps/dashboards/APIs. Act supports inspection and planning.

4. Hardware components

Camera: captures visual road evidence. IMU/accelerometer: captures vibration and impact patterns. GNSS/NavIC receiver: supplies position and time. Temperature sensor: adds environmental context. ESP32: sensor acquisition, timestamps and basic filtering. Edge computer: local vision, event logic, storage and sensor fusion. Communication/power/storage complete the prototype.

5. Why use multiple components?

A camera can see a defect but can be affected by lighting and viewpoint. IMU can sense roughness but is affected by vehicle dynamics. Location alone cannot detect a defect. Combining independent signals allows cross-checking, confidence estimation and repeated segment-level evidence.

6. Electronics integration

Sensors feed the ESP32 and/or suitable interfaces. The ESP32 timestamps and packages readings. The edge computer processes images, detects candidate events, combines sensor evidence and stores records. A suitable communication link transfers selected records to the platform. The exact hardware interface should be finalized during prototyping.

7. NavIC

NavIC (Navigation with Indian Constellation) is India's independent regional navigation system developed by ISRO. Its primary service area covers India and about 1,500 km beyond the Indian mainland. It provides positioning, navigation and timing. In ROAD-SENSE, NavIC/GNSS is the spatial and temporal reference: it tells the system where and when an observation occurred.

8. Current NavIC technical points

ISRO's recent annual report describes a base layer currently providing PNT service through four satellites, with NVS-02 providing limited PNT service after its 2025 launch. The NVS second-generation series is intended to augment NavIC and adds L1 capability. NavIC's ground segment is fully functional, with continuing work on reference stations and related capabilities.

9. NISAR and ISRO ecosystem

NISAR is a joint NASA-ISRO Earth-observation mission. It uses dual-frequency L-band and S-band synthetic aperture radar, a 12 m reflector, and a sun-synchronous orbit around 747 km with a 12-day repeat cycle. NISAR

is not a navigation system. In the ROAD-SENSE concept, Earth-observation data could provide broader environmental or land-surface context while ground sensors provide road-level evidence.

10. End-to-end workflow

Capture → Locate → Pre-process → Detect → Fuse → Confidence → Aggregate by road segment → Map/Score/Alert → Inspection priority. This workflow is the central technical story of the project.

11. Road-health intelligence

A proposed health metric can combine anomaly frequency, visual severity, vibration correlation, repeated observations, location consistency and environmental context. Confidence should reflect agreement and repeatability. Example values such as HEALTH 42/100 or CONFIDENCE 94% are illustrative unless measured in a real validation study.

12. Key benefits

Continuous repeated evidence, multi-sensor cross-checking, location-aware records, visible uncertainty, edge processing, road-segment history, deterioration trends and decision-ready outputs.

13. Long-term perspective

Stage 1: controlled prototype. Stage 2: supervised pilot with repeated routes. Stage 3: multi-fleet road-health network. Scaling should follow technical validation, privacy review, safety controls and engineering verification.

14. Validation

Test detection accuracy on known controlled conditions; localization correctness; repeatability across passes; whether sensor fusion reduces false alarms; processing latency; and confidence calibration. Do not present illustrative numbers as field performance.

15. Entrepreneurship

Potential users include municipalities, road agencies, fleet operators, researchers and mapping/mobility platforms. Possible models include pilot deployments, fleet subscriptions, analytics/API services, institutional dashboards and research partnerships.

16. One-minute pitch

ROAD-SENSE uses a camera to see road conditions, an IMU to measure vibration, temperature for context and NavIC/GNSS for position and time. ESP32 handles sensor acquisition while an edge computer processes and fuses the evidence. Repeated observations become road-segment health, confidence and trend information. The goal is to create a road-health intelligence layer that complements existing maps and supports smarter inspection and maintenance decisions.

Official ISRO facts used for the satellite sections

- ISRO — NavIC / Navigation Systems, Annual Report 2025–26.
- ISRO — NVS-02: Advancing India's Regional Navigation Capabilities.
- ISRO — NISAR Mission page and NISAR S-Band SAR Data Products Release (2026).
- All ROAD-SENSE architecture, scoring, business model and performance examples are proposed project elements unless explicitly identified as official facts.